Product Inventory Stock-Level Monitor
Excel project report. Data snapshot: 27 October 2026

1. Objective

- I built an Excel monitor that shows which products are low or out of stock, so nothing runs out and nothing piles up. My motto: “Never run out, never overstock.”
- It answers four questions without manual filtering: what to reorder, how much, from which supplier, and how much money is tied up in stock.

2. Business problem

- With 200 products, checking stock by eye is slow and risky.
- A stockout means lost sales. Overstock means cash stuck on shelves.
- This project turns a stock list into a monitor that updates itself when the numbers change.

3. Dataset

- 200 products, 15 suppliers, and 7 categories: Books & Media, Electronics, Grocery, Home & Kitchen, Personal Care, Sports & Fitness, Stationery.
- The data is synthetic. I created it for this project, and there is no sales or demand history.
- Seven columns are raw: Product ID, Product Name, Category, Supplier ID, Quantity in Stock, Reorder Level, Last Restocked Date.
- Seven columns are formulas: Supplier, Unit Price, Stock Value, Lead Time, Stock Status, Reorder Quantity, Days Since Restock.
- Two lookup sheets support them: `Supplier_Master` and `Product_Master`.
- I added mistakes on purpose (duplicate rows, extra spaces, wrong letter case) so I could show data cleaning.

4. Method

- **Cleaning:** TRIM, UPPER, PROPER, and Remove Duplicates, so lookups match exactly.
- **Validation:** dropdowns for Category and Supplier ID, and a whole-number rule for quantities.
- **Lookups:** XLOOKUP for supplier name and lead time, and a lookup for unit price from `Product_Master`.
- **Logic:** IFS for stock status, IF for reorder quantity, IFERROR for stock value, and DATEDIF with TODAY for days since restock.
- **Summaries:** COUNTIFS, SUMIFS, and AVERAGEIFS by category and by supplier, plus five alert tiles. INDEX-MATCH finds the supplier with the most alerts.
- **Checks:** five check cells on the `Summary` sheet that must all equal 0. They do.
- **Visuals:** conditional formatting, three PivotTables, charts, and slicers.
- **Dashboard:** KPI cards built with named ranges, and a reorder list built with FILTER.

Business rules I used

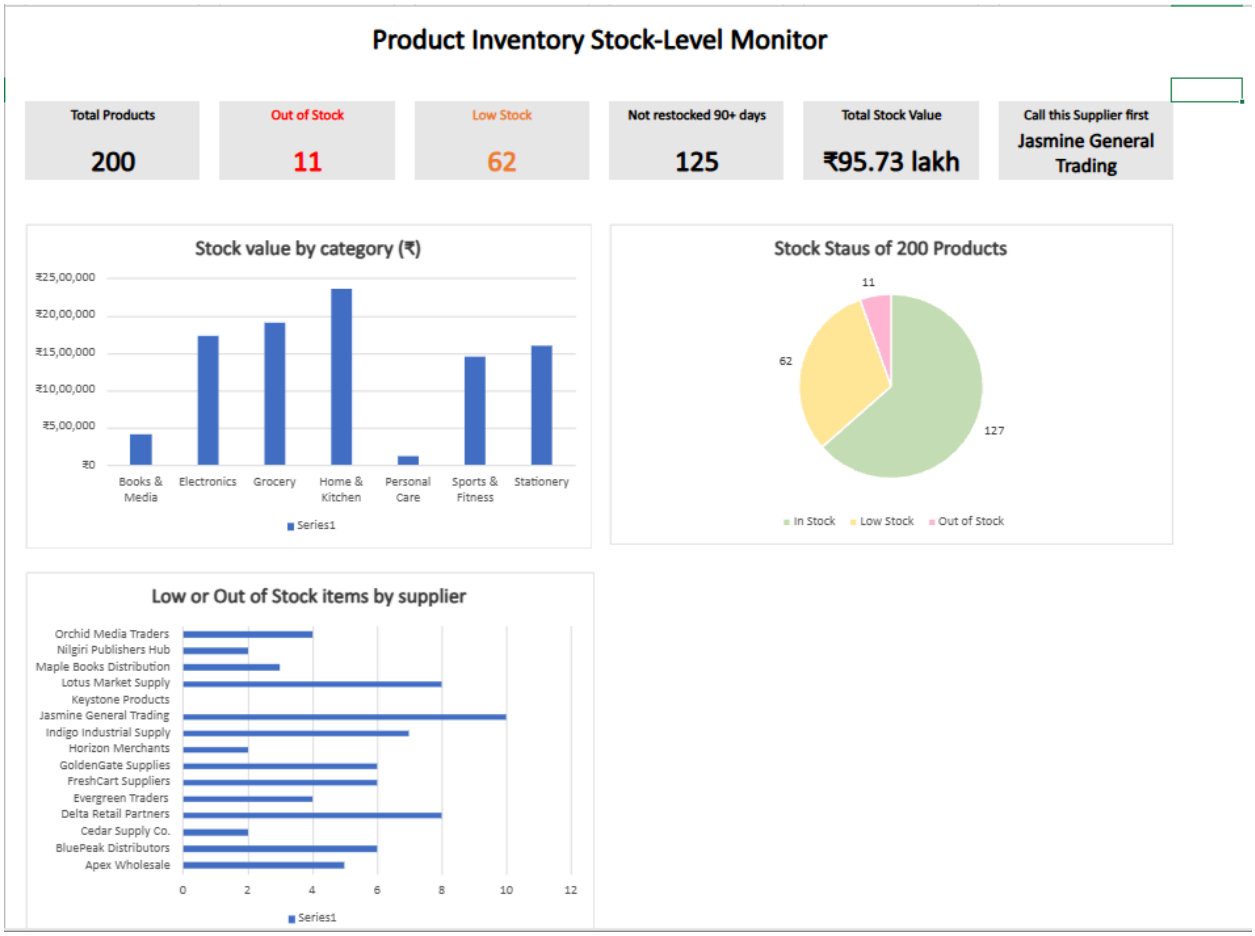
- Quantity 0 is Out of Stock. Quantity below the Reorder Level is Low Stock. Anything else is In Stock.
- Reorder Quantity means “order up to twice the Reorder Level”: (2 × Reorder Level) − Quantity. It is 0 if the product is In Stock.
- A product whose quantity equals its Reorder Level counts as In Stock, because the rule uses “below”.

5. Data cleaning results

- I started with 203 rows. Remove Duplicates took out 3, leaving 200.
- I fixed 4 product names with extra spaces and 3 category names with the wrong case (check these counts against your `Data_Issues_Log`).
- Product names only got TRIM. PROPER would have damaged names such as “USB-C Cable” and “Basmati Rice 5kg”.
- After cleaning, my checks showed zero problems.

6. Dashboard

- Dashboard and Summary screenshot are below for reference



Inventory Summary							
Category	Products	Total Quantity	Total Stock Value	Avg Unit Price	Out of Stock	Low Stock	In Stock
Books & Media	25	1,233	₹4,11,889	₹321.48	2	7	16
Electronics	35	1,422	₹17,23,478	₹1,061.86	2	23	10
Grocery	35	10,445	₹19,01,162	₹212.66	1	4	30
Home & Kitchen	30	4,361	₹23,64,989	₹531.67	2	3	25
Personal Care	24	498	₹1,10,602	₹247.33	2	20	2
Sports & Fitness	23	2,782	₹14,57,518	₹535.96	1	4	18
Stationery	28	10,440	₹16,03,024	₹139.93	1	1	26
Total	200	31,181	₹95,72,662	₹453.88	11	62	127

Supplier Summary		
Supplier	Low or Out Items	Stock Vlaue
Apex Wholesale	5	₹1,69,715
BluePeak Distributors	6	₹3,68,786
Cedar Supply Co.	2	₹8,11,668
Delta Retail Partners	8	₹7,36,045
Evergreen Traders	4	₹4,40,056
FreshCart Suppliers	6	₹7,86,055
GoldenGate Supplies	6	₹10,95,025
Horizon Merchants	2	₹11,66,714
Indigo Industrial Supply	7	₹10,26,269
Jasmine General Trading	10	₹5,86,881
Keystone Products	0	₹14,34,678
Lotus Market Supply	8	₹5,38,881
Maple Books Distribution	3	₹1,29,614
Nilgiri Publishers Hub	2	₹1,34,962
Orchid Media Traders	4	₹1,47,313
Total	73	₹95,72,662

Label	Value
Products Out of Stock	11
Products Low Stock	62
Not restocked in over 90 days	125
Total stock value	₹95,72,662
Supplier with most Low or Out items	Jasmine General Trading
Products vs data rows	0
Statuses add up	0
Category value total	0
Supplier value total	0
Supplier alerts vs categories	0

- The KPI cards show: 200 products, 11 out of stock, 62 low stock, 125 not restocked in 90+ days, ₹95.73 lakh total stock value, and the supplier to call first (Jasmine General Trading).
- The charts show stock value by category, stock status, and low or out items by supplier.

- `Reorder_List` uses the FILTER function to show every product whose Reorder Quantity is above zero (73 products at the snapshot date), with supplier and suggested order quantity. It updates automatically when stock data changes.
- The slicers are on the `Pivot` sheet and filter PivotTable 3 only.

7. Insights

- 73 of 200 products (36.5%) need reordering now: 11 are Out of Stock and 62 are Low Stock.
- Personal Care has 22 of 24 products Low or Out. Electronics has 25 of 35. Together they make 47 of the 73 alerts.
- Home & Kitchen (₹23,64,989), Grocery (₹19,01,162), and Electronics (₹17,23,478) hold about 63% of the ₹95,72,662 stock value.
- Grocery and Stationery hold 67% of all units (20,885) but only 37% of the value (₹35,04,186). They are high volume and low price.
- Four suppliers cause 45% of the alerts: Jasmine General Trading (10), Delta Retail Partners (8), Lotus Market Supply (8), and Indigo Industrial Supply (7).
- Indigo (8 days) and Delta (7 days) have the longest delivery times, which raises their stockout risk.
- 125 of 200 products (62.5%) have not been restocked in 90+ days. This number grows every day.
- Keystone Products has no alerts and holds ₹14,34,678 of stock, so it works as a benchmark for a healthy supplier.

8. Recommendations

- Place orders now for the 73 products on `Reorder_List`, starting with the 11 Out of Stock.
- Look at Personal Care and Electronics first, and review whether their reorder levels are too low.
- Contact Jasmine, Delta, and Lotus first, and consider combining orders.
- Before reordering Grocery or Stationery, check for overstock, and set an overstock rule (for example, quantity above 5 times the reorder level).
- Refresh the PivotTables and review the `Summary` checks every week.

9. Limitations and next steps

- The data is synthetic, and there is no sales history, so the reorder levels are assumptions.
- The reorder rule (order up to twice the reorder level) is my own simple rule and ignores each supplier’s standard order quantity.
- I used Excel Online, so there are no VBA macros. Slicers stay on the `Pivot` sheet and filter PivotTable 3 only.
- Days Since Restock changes every day because it uses TODAY().

- **Next steps:** add VBA macros for refresh and reset, and connect slicers to every PivotTable in desktop Excel.

Appendix: the formula list from the `Formula_Log` sheet.